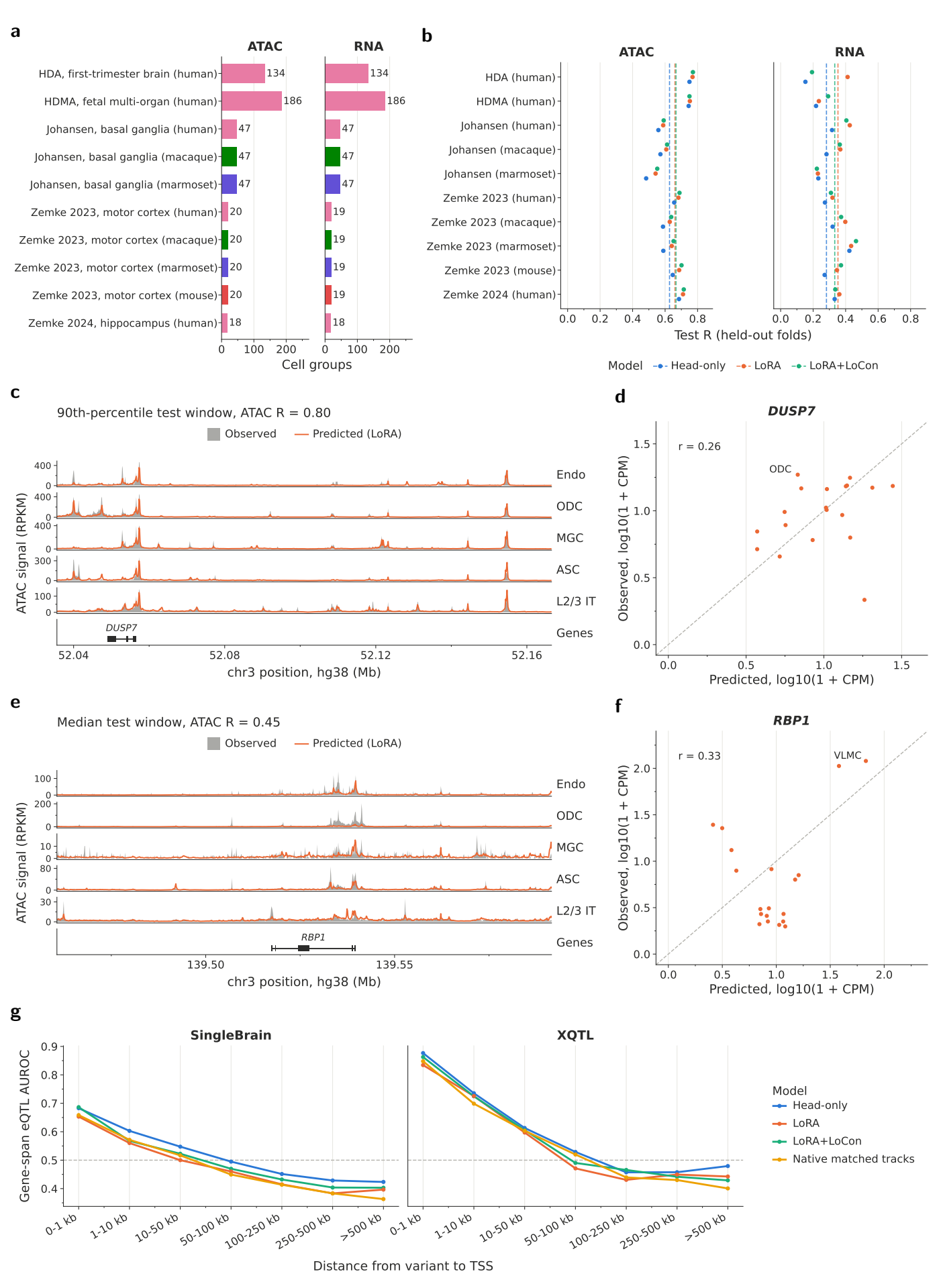

a
ATAC
RNA
HDA, first-trimester brain (human)
HDMA, fetal multi-organ (human)
Johansen, basal ganglia (human)
Johansen, basal ganglia (macaque)
Johansen, basal ganglia (marmoset)
Zemke 2023, motor cortex (human)
Zemke 2023, motor cortex (macaque)
Zemke 2023, motor cortex (marmoset)
Zemke 2023, motor cortex (mouse)
Zemke 2024, hippocampus (human)
134
186
47
47
47
20
20
20
20
134
186
47
47
47
19
19
19
19
18
18
0 100 200
0 100 200
Cell groups
b
ATAC
RNA
HDA (human)
HDMA (human)
Johansen (human)
Johansen (macaque)
Johansen (marmoset)
Zemke 2023 (human)
Zemke 2023 (macaque)
Zemke 2023 (marmoset)
Zemke 2023 (mouse)
Zemke 2024 (human)
0.0 0.2 0.4 0.6 0.8
0.0 0.2 0.4 0.6 0.8
Test R (held-out folds)
Model
Head-only
LoRA
LoRA+LoCon
c
90th-percentile test window, ATAC R = 0.80
Observed
Predicted (LoRA)
ATAC signal (RPKM)
Endo
ODC
MGC
ASC
L2/3 IT
Genes
DUSP7
52.04
52.08
52.12
52.16
chr3 position, hg38 (Mb)
d
DUSP7
r = 0.26
ODC
Observed, log10(1 + CPM)
Predicted, log10(1 + CPM)
e
Median test window, ATAC R = 0.45
Observed
Predicted (LoRA)
ATAC signal (RPKM)
Endo
ODC
MGC
ASC
L2/3 IT
Genes
RBP1
139.50
139.55
chr3 position, hg38 (Mb)
f
RBP1
r = 0.33
VLMC
Observed, log10(1 + CPM)
Predicted, log10(1 + CPM)
g
SingleBrain
XQTL
Gene-span eQTL AUROC
0-1 kb
1-10 kb
10-50 kb
50-100 kb
100-250 kb
250-500 kb
>500 kb
Distance from variant to TSS
Model
Head-only
LoRA
LoRA+LoCon
Native matched tracks